# CLOCK DIVIDER CIRCUIT DESIGN USING VERILOG

**Multi-frequency clock divider**

Submitted by

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## 1 Introduction

Clock division is an essential operation in digital systems where different components require different clock frequencies. A clock divider reduces the frequency of a high-speed input clock to lower frequencies suitable for various applications.

In this project, a simple and efficient clock divider is designed using Verilog by employing a 3-bit binary counter. The design generates output clocks divided by 2, 4, and 8, which are verified through simulation.

## 2 Design Explanation

The clock divider circuit is designed to generate output clock signals with frequencies divided by 2, 4, and 8 from a single input clock. This is achieved using a 3-bit binary counter.

The counter increments on every rising edge of the input clock. Each bit of the counter toggles at half the frequency of the previous bit. Thus:

- Bit 0 produces a clock divided by 2
- Bit 1 produces a clock divided by 4
- Bit 2 produces a clock divided by 8

By directly assigning these bits to outputs, multiple divided clock signals are generated efficiently.

# 3 Module Description

## 3.1 Top Module (clock_divider_top)

This module serves as the main integration unit of the design. It instantiates the 3-bit counter and connects its outputs to the divided clock signals. The input clock and reset signals are provided to the counter, and the counter outputs are mapped directly to the required clock divisions.

**Code Snippet:**

```
assign clk_div2 = count[0];
assign clk_div4 = count[1];
assign clk_div8 = count[2];
```

## 3.2 Counter Module (counter_3bit)

This module implements a 3-bit synchronous counter. It increments its value on every rising edge of the clock signal. When the reset signal is active, the counter is reset to zero. The changing binary values of the counter produce the required frequency-divided outputs.

**Code Snippet:**

```
always @(posedge clk or posedge rst) begin
    if (rst)
        count <= 3'b000;
    else
        count <= count + 1;
end
```

## 3.3 Testbench Description

The testbench is used to verify the functionality of the clock divider circuit. A clock signal with a time period of 10 ns is generated using a toggling mechanism. The reset signal is initially asserted to initialize the counter and then de-asserted to allow normal operation. The output waveforms are observed to confirm correct frequency division.

**Clock Generation:**

```
always #5 clk = ~clk;
```

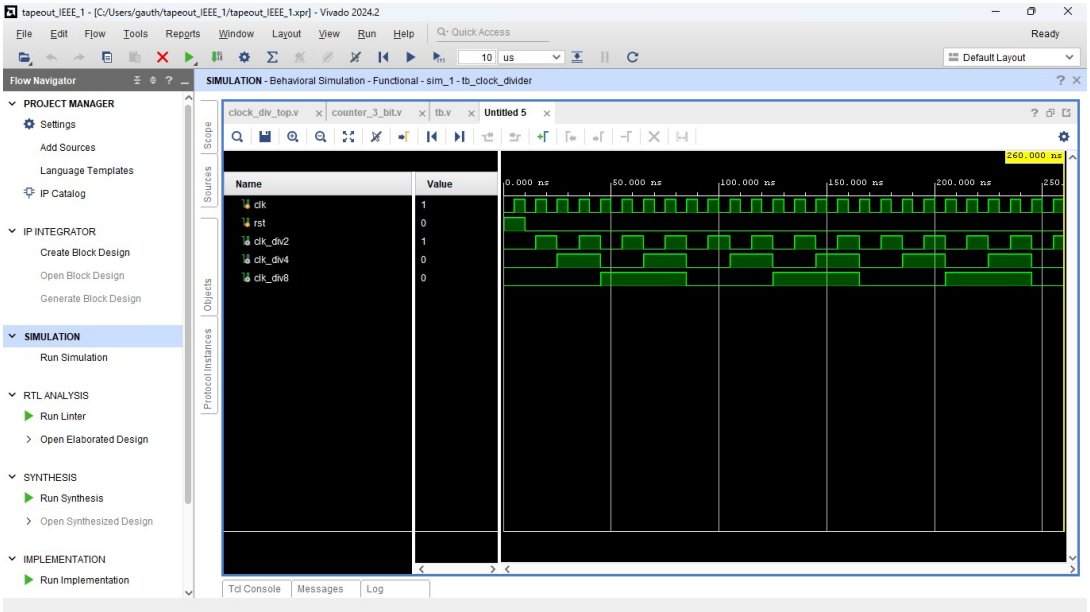
# 4 Results

The designed clock divider circuit was simulated using a Verilog testbench. The output waveforms confirm correct frequency division:

- clk_div2 operates at half the input clock frequency

- clk_div4 operates at one-fourth the input clock frequency
- clk_div8 operates at one-eighth the input clock frequency

**Simulation Waveform:**




# 5 Challenges Faced

- Understanding how binary counter bits correspond to different frequency divisions.
- Ensuring correct reset operation for proper initialization.
- Observing and verifying frequency changes in waveform simulation.
- Debugging initial issues in clock generation within the testbench.

# 6 Conclusion

The clock divider circuit was successfully designed and implemented using Verilog. By utilizing a 3-bit counter, output clocks with frequencies divided by 2, 4, and 8 were generated efficiently. The simulation results verified the correct operation of the design, demonstrating the effectiveness of using counters for frequency division.